CERNEA BOGDAN-ANDREI

Full-Stack Developer | Computer Science Student

(+40) 771 511 797 | cerneabogdanandrei@gmail.com | linkedin.com/in/cernea-bogdan-andrei | github.com/CerneaBogdanAndrei | cerneabogdanandrei.github.io/Cv | Galati, Romania

PROFESSIONAL PROFILE

Full-Stack Developer with proven experience building scalable web applications using TypeScript, NestJS and React. Hands-on expertise in relational database architecture, Docker containerisation, real-time systems, machine learning and computer vision. Final-year Computer Science & IT student at "Dunarea de Jos" University (Galati, Romania), with a strong portfolio of complex projects, real-world internships and 1st-place finishes in AI cyber-defence and industrial robotics hackathons.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, Java, PHP, SQL
Frontend	React, Next.js, WebGL, Three.js, HTML5, CSS3
Backend	NestJS, Node.js, Prisma ORM, REST APIs, PHP (PDO)
Databases	PostgreSQL, Supabase, Firebase, MySQL
DevOps	Docker, Linux, Git, LDAP, CI/CD
ML / AI	TensorFlow, scikit-learn, OpenCV (EfficientNetV2, image & network-traffic classification, computer vision)
Tools	VS Code, IntelliJ IDEA (full JetBrains ecosystem), Google Antigravity, XAMPP, Retool, Figma

PROFESSIONAL EXPERIENCE

IT Technician & Web Developer (part-time) | GlobalECR CRIS SRL — *Mar 2026 – Aug 2026*

- Configuration and installation of fiscal cash registers; connectivity and troubleshooting for ANAF government servers.
- Development and modernisation of globalecr.ro with fluid animations, responsive design and intuitive UX, using React and TypeScript.
- Direct team collaboration for the iterative delivery of new features.

Tech: *React, TypeScript, HTML/CSS*

Internship – Full-Stack Developer | WindRiver Systems — *2026*

- Developed a full-stack academic timetable application for the faculty as part of a 5-person team.
- Built the frontend in React and TypeScript; implemented API connectivity and student account sorting using Python.
- Implemented NestJS modules with Prisma ORM for academic data management.
- Set up Docker infrastructure and integrated LDAP authentication with the university identity system.
- Collaborated using Git workflows (branch strategy, pull requests, code review).

Tech: *TypeScript, React, Python, NestJS, Prisma ORM, PostgreSQL, Docker, LDAP, Git*

Internship – Web Developer | APAN — *2025*

- Development and styling of dynamic web pages in a real production environment.

Tech: *HTML, CSS, PHP*

Academic Internship – Developer | Faculty ACIEE — *2025*

- Individual project: Java desktop application for managing a car repair service (clients, vehicles, bookings, interventions).
- MVC architecture with DAO classes and MySQL connectivity via JDBC.
- Role-based authentication and dynamic UI (admin, owner, employee), with a switchable Dark Mode.

Tech: *Java, Swing, JDBC, MySQL, NetBeans*

RELEVANT PROJECTS

Autonomous Robotic Arm – DOFBOT — *Team project — 1st Place, CRI 2026 Industrial Robots Hackathon*

Python · OpenCV · Raspberry Pi 4 · Yahboom DOFBOT · Camera

- Recognises the coloured cubes on a 3×3 grid (row, column, top-face colour), memorises the arrangement and reproduces it on a second grid; builds colour-ordered towers at a jury-chosen cell.
- My role: the AI / computer-vision part (cube and grid detection, scene memory, move planning), integrated with the team's arm-motion code.

Voxel Engine – Browser Minecraft Clone — *Personal project — ongoing*

TypeScript · WebGL · Three.js

- Voxel-based graphics engine running natively in the browser, with strict OOP architecture and clear separation of concerns.
- Efficient memory management: 3D structures converted to linear Uint8Array buffers with O(1) access.
- Render optimisation via face culling of invisible geometry, significantly reducing draw call count.

AI Medicinal Plant Identification System — *Team project — 2nd Mention, Severin Bumbaru Creativity Contest*

Python · TensorFlow · EfficientNetV2 · TypeScript · Supabase — Android (.apk) & Windows (.exe)

- End-to-end development of an application identifying 15+ medicinal plant species through image processing.
- Cross-platform architecture delivered and tested on both Android and Windows desktop.
- Full database design and backend infrastructure integration using Supabase.

GurmandFit – Nutrition Blog Modernisation — *Personal project — ongoing*

TypeScript · React · PostgreSQL · Supabase

- Full migration from WordPress to a custom full-stack architecture built from scratch in TypeScript.
- Rich-text post editor similar to Microsoft Word, integrated directly into the site interface.
- Structured database with dedicated modules for recipes, workouts and nutritional content.

Auto Dealership Management Portal (Multi-role) — *Team project*

Retool · Supabase · JavaScript

- Multi-role system (sales, technical, IT, mechanic) with real-time task routing: bookings are automatically distributed to the appropriate role.
- Authentication with JS logic differentiating employees from clients via separate database lookups.

Insurance Company Portal — *Academic project*

React · TypeScript · PHP (PDO) · MySQL

- Complete web solution for managing contracts and client data, with PHP backend and relational data storage.

EDUCATION

Computer Science & Information Technology (CTI) | "Dunarea de Jos" University of Galati — ACIEE *2023 – present | Year IV*

COMPETITIONS & AWARDS

- **1st Place — AI Cyber Defense Challenge, ZTS XIII (OSACE, University of Craiova, Nov 2026):** team ML hackathon — models that detect cyberattacks in network traffic, classify attack types and flag unknown attacks (Macro F1, AUROC).
- **1st Place — CRI 2026 Industrial Robots Hackathon (Galati):** autonomous DOFBOT robotic arm (Raspberry Pi 4 + camera) that memorises and reproduces a 3×3 arrangement of coloured cubes and builds colour-sorted towers; I built the AI / computer-vision part. Code: github.com/CerneaBogdanAndrei/DYD-concurs-robotica.
- **1st Mention — E.ON Energy Challenge (national, team Future Energy Solutions):** hydrogen-based cogeneration station — research and prototyping; argued the need for an anti-flashback valve over the bubbling system and presented the solution to the national jury.
- **2nd Mention — Severin Bumbaru Creativity Contest:** AI Medicinal Plant Identification System (see Projects).
- **54th of 296 ranked teams — CSCTF 2026: Total Warfare, Chronos Security (Nov 2026):** 48-hour international online CTF (Fighters division, team S&B Software), 2,194 participants — attacks on energy, telecom, healthcare, finance and defence systems.
- **3rd Place — CyberSea Fest CTF, organised by Chronos Security (2026):** national cybersecurity competition — OSINT, cryptography, web exploitation, Windows/Linux challenges.

LANGUAGES

Romanian	Native
English	B2

CERTIFICATIONS

- **Elements of AI — University of Helsinki / MinnaLearn** (2024): 2 ECTS credit course, fundamentals of AI and machine learning.
- **Digital Skills Wallet — EU Micro-credentials** (2026): AI in search & social media, misinformation/bias in AI content, data collection, and data mining fundamentals.
- **Other courses:** PC Operator / MS Office (2022), Entrepreneurship (2022), Social Entrepreneurship (2025) — Social Trade; Financial-Accounting Entrepreneurship — UGAL.

ADDITIONAL EXPERIENCE

- **AIESEC Galati — Volunteer** (Nov 2023 – Feb 2024): international volunteering campaigns, cross-cultural communication.
- **Assistant Manager — Access Visual Art Galati** (2019–2022): course supervision, client relationship management.